National University of Computer and Emerging Sciences

Information Retrieval (CS4051)

Date: April 09,2026

Course Instructor

Muhammad Rafi

Sessional-II Exam

Total Time: 1 Hour

Total Marks: 40

Total Questions: 03

Semester: Spring-2026

Campus: Karachi

Department: AI & DS

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CLO # 1: < Understand the basic concepts and techniques in Information Retrieval.>

Q1. Answer the following questions to the point. **[Time: 30 mins] [Marks: 2x10]**

a. What are the three important components of information retrieval system evaluation?

Three important components of Information Retrieval (IR) system evaluation are:

1. Document collection - The set of documents over which retrieval is performed.
2. Queries / information needs -The user requests or search topics used to test the system.
3. Relevance judgments - A ground-truth indication of which documents are relevant or non-relevant for each query.

b. How the idea of inexact K document retrieval expedites the processing of search results?

Inexact Top-K retrieval speeds up search by avoiding scoring all documents exactly. Instead, the system tries to quickly find a set of documents that are very likely to be among the top K results. This expedites processing because of (1) fewer documents are fully scored, (2) approximate methods reduce computation, and (3) users usually care only about the top few results, not the complete ranking. So, the system trades a little exactness for much faster response time.

c. What is a Champion list?

A Champion list is a precomputed list of the top-ranked documents for a term, usually based on a score such as term frequency or impact. For each term, the system stores only a small set of "best" documents. During retrieval, it searches these preferred documents first instead of the full postings list. This makes retrieval faster.

d. What is query expansion? Suggest two methods of global query expansion.

Query expansion means adding extra related terms to the original query so that more relevant documents can be retrieved. Two methods of global query expansion are (1) Using a thesaurus. and (2) Using term associations / co-occurrence statistics. Global methods are called "global" because they use collection-wide information, not just top retrieved documents.

e. What are static search snippets? Why these are not good for relevance feedback?

Static search snippets are fixed short summaries prepared in advance for documents, often independent of the current query (usually first 100 bytes from the documents). They are not good for relevance feedback because (1) they may not highlight the part of the document relevant to the current query, (2) they can omit useful query-related evidence and (3) feedback based on them may be misleading.

f. Can Rocchio completely change the user's original query meaning? Under what parameter settings might this happen? (hint: α, β and γ)

Yes, Rocchio can significantly alter or even effectively change the original query meaning if the influence of feedback dominates the original query.
This may happen when (1) α is very small, so the original query has little weight, (2) β is large, so relevant documents strongly pull the query, and (3) γ is also large, so non-relevant documents strongly push the query away from certain terms. So query drift is more likely when low α, high β, and/or high γ are used.

g. In Probability Ranking Principle (PRP), why there is an assumption for null document i.e. $P[d=0/ R=1, q] = P[d=0/ R=0, q] = 0$? what it signifies?

This assumption means the null document (No real feature) has zero probability under both relevance and non-relevance. It signifies that only real documents in the collection are considered in ranking. A non-existent document cannot meaningfully be treated as relevant or non-relevant.

h. What is meant by a multinomial distribution over words in the context of information retrieval?

A multinomial distribution over words means a document is modeled as a probability distribution over vocabulary terms. Words in the document are assumed to be generated by repeated draws from that distribution. This idea is central in language modeling approaches to IR.

i. List any two limitations of the Normalized Discount Cumulative Gain (NDGC)?

First, NDCG depends on graded relevance judgments, which are costly and often subjective. Second, it is sensitive to the choice of gain values and discount function, so scores may vary by setup. Thus, interpretation is not always straightforward.

j. What is the value of NDCG for a perfect ranking algorithm?

For a perfect ranking, NDCG = 1. This is because the obtained ranking is identical to the ideal ranking, so DCG = IDCG.

CLO # 2: <Understanding how IR Model works >

Q2. Consider the following documents in a collection. **[Time: 15 mins] [Marks: 10]**

Vocabulary = {w1 w2 w3 w4 w5 w6}, where the number indicates the order of dimensions:

d1= {w3 w2 w5}
d2= {w3 w2 w6 w1}
d3= {w1 w2 w3}
d4= {w6 w4 w5}

Relevant Documents = d1, d3, d4
Non-Relevant Document= d2
q = {w2 w4 w5}

Using the Probability Ranking Principle (PRP) to rank these documents. You can use the Laplace smoothing version as below:

$N_1 = P(w_i \mid R) = (count\ (w_i\ in\ R) + 1) / (|R| + |V|)$
$N_0 = P(w_i \mid \neg R) = (count\ (w_i\ in\ \neg R) + 1) / (|\neg R| + |V|)$

V = { w1,w2,w3,w4,w5,w6} and |V| = 6
R= { d1,d3, d4} and |R|= 3
~R ={d2} = |~R| = 1
Q = {w2,w4,w5}

	w1	w2	w3	w4	w5	w6
N_1	(1+1/3+6)=2/9	1/3	1/3	2/9	1/3	1/9
N_0	(1+1/ 1+6)=2/7	2/7	2/7	1/7	1/7	2/7

For probability ranking the document given the query q=(w2 w4 w5). We will have

P(wi/R)	P(wi/~R)
P(w2/R)=1/3 P(w4/R) = 2/9 P(w5/R) = 1/3	P(w2/~R) = 2/7 P(w4/~R)= 1/7 P(w5/~R) = 1/7

We know,

$$c_i = \log \frac{P(w_i \mid R)\,[1 - P(w_i \mid \neg R)]}{P(w_i \mid \neg R)\,[1 - P(w_i \mid R)]}$$

So, c_{w2} = log (6/4); c_{w4}= log (12/7) and c_{w5}= log (3)

Now RVS(d1) , it contains only 2 terms form 3 query terms, hence
RVS(d1) = $c_{w2} + c_{w5}$ = log(15/4) =1.322
RVS(d2) = log(5/4) =0.223
RVS(d3) = log(5/4) =0.223
RVS(d4) = log(36/7) =1.638

Hence ranking d4>d1>d3=d4

CLO # 3: < Appreciate the importance of data structures, indexes and retrieval efficiency > (part a)
CLO # 4 < Appreciate the importance of data structures, indexes and retrieval efficiency> (part b)

Q3. Answer the following questions. **[Time: 15 mins] [Marks: 2x5]**

a. Compare and contrast three main differences in Probabilistic IR vs Language Modeling for IR

Probabilistic Model	**Language Model**
Probabilistic IR ranks documents by estimating the probability of relevance, that is, the likelihood that a document is relevant to a query, commonly viewed as P(R=1|q,d).	Language Modeling for IR ranks documents by estimating how likely the document model is to generate the query, that is, P(q|Md).
In Probabilistic IR, relevance is an explicit part of the model. Harder to calculate from the given data.	In Language Modeling, relevance is handled indirectly through query generation probability rather than being modeled as a separate variable.
Probabilistic IR focuses on relevance-based matching between query and document.	Language Modeling treats each document as a language source and asks whether the query could have been generated from that document's language model. This gives a generative interpretation of retrieval.

b. Consider the following two documents

d1: cricket team wins final match with strong batting
d2: football team loses final game after poor defense

Let λ = 1/2 and Suppose the query is: "**final team wins**" estimate the following:

(i) Collection Probabilities for the query terms.

P(final/M_c) = 1/8; P(team/M_c) = 1/8; P(wins/M_c) = 1/16

(ii) MLE document probabilities with document model (M_d)

For document d1: $P(final/M_{d1}) = 1/8$; $P(team/M_{d1}) = 1/8$; $P(wins/M_{d1}) = 1/8$

For document d2: $P(final/M_{d2}) = 1/8$; $P(team/M_{d2}) = 1/8$; $P(wins/M_{d2}) = 0$

(iii) Query likelihood estimate for the given document using mix mode with given λ

The formula is given as below:

$$\hat{P}(t|d) = \lambda \hat{P}_{\text{mle}}(t|M_d) + (1-\lambda)\hat{P}_{\text{mle}}(t|M_c)$$

Where M_d and M_c have the same meaning we discussed during the lectures.

Using Query likelihood estimates for D1:

P(final/d1) = 1/2 X 1/8 + 1/2 X 1/8 = 1/8
P(team/ d1) = 1/2 X 1/8 + 1/2 X 1/8 = 1/8
P(wins/d1) = 1/2 X 1/8 + 1/2 X 1/16 = 3/32

Now P(q/d1) = 1/8 X 1/8 X3/32 = 3/2048

P(final/d2) = 1/2 X 1/8 + 1/2 X 1/8 = 1/8
P(team/ d2) = 1/2 X 1/8 + 1/2 X 1/8 = 1/8
P(wins/d2) = 1/2 X 0 + 1/2 X 1/16 = 1/32

Now P(q/d2) = 1/8 X 1/8 X1/32 = 1/2048

As P(q/d1) > P(q/d2) – document d1 is more relevant to the query, under query likelihood model.

<The End.>